Firms in Competitive Markets

EC 111 Principles of Microeconomics

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Reading

- These slides are not a substitute for reading Mankiw, Chapter 15, "Firms in Competitive Markets".
- Required reading: Mankiw, *Principles of Microeconomics*, 11th edition, Chapter 15, "Firms in Competitive Markets".

What you must know from this chapter

1. Overview of market structures.
2. Conditions necessary to achieve perfect competition.
3. Profit maximization (MR = MC).
4. Graphical representation of profits/losses.
5. Shut-down point – short-run and long-run.
6. Short-run supply curve.
7. Entry/exit.
8. Firm and industry equilibrium and changes to equilibrium.

Part 1

Market Structures and Perfect Competition

Four market structures

Definition: Market structure

The number of firms, whether products are identical or differentiated, and how easy it is to enter or leave the market.

Structure	Main feature	Firm's pricing position
Perfect competition	Many sellers of identical goods	Price taker
Monopolistic competition	Many sellers of differentiated goods	Some say over price
Oligopoly	A few important competing firms	Decisions depend on rivals
Monopoly	One seller; no close substitutes	Faces market demand

Perfect competition

Definition: Perfectly competitive market

- Many buyers and many sellers.
- Firms sell **identical** products.
- Each buyer and seller is too small to move the market price.
- Firms can enter and leave freely in the long run.

Free entry requires no significant barriers, including legal barriers that prevent new firms from entering.

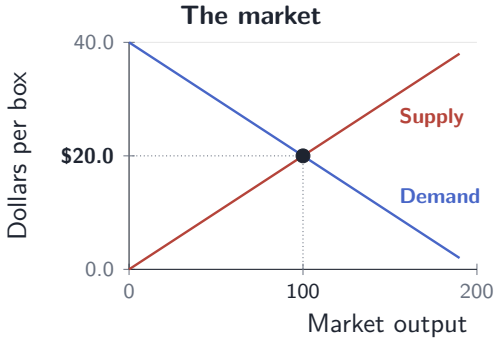
Price taker

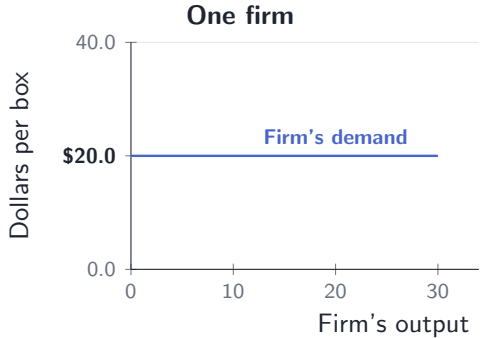
Definition: Price taker

A buyer or seller who accepts the market price and does not choose it.

- Identical boxes of labels sell for $20.0 in the market.
- A seller asking $23.0 loses its buyers to sellers charging $20.0.
- Selling another box does not require a price cut.
- The firm chooses its output. A change in its own output does not change the market price.

Market demand versus firm demand





Demand: $P = 40 - 0.2Q$. Supply: $P = 0.2Q$. At $Q = 100$: $40 - 0.2(100) = 0.2(100) = \$20.0$.

Finding

Market demand slopes down. One firm's demand is horizontal at the market price. A change in the market price moves the firm's demand line.

Part 2

Profit Maximization: MR = MC

The label printer

- The printer makes standard boxes of shipping labels. Q is boxes per day.
- Fixed cost is \$160.0 per day, including unavoidable lease payments.
- Total cost: $\text{TC} = 0.02Q^3 - 0.4Q^2 + 6Q + 160$.
- Variable cost: $\text{VC} = \text{TC} - 160$.
- The printer takes the market price. It chooses how many boxes to make.

Revenue and economic profit

Definition: Total, average and marginal revenue

Total revenue: $\text{TR} = P \times Q$.
Average revenue: $\text{AR} = \text{TR} \div Q$, for $Q > 0$.
Marginal revenue: $\text{MR} = \Delta\text{TR}/\Delta Q$.

Definition: Economic profit

$\pi = \text{TR} - \text{TC}$. Total cost includes explicit payments and opportunity costs.

When output rises one box at a time, $\Delta Q = 1$. MR is the extra revenue from that box.

Price, average revenue and marginal revenue

The market price is \$23.5 per box.

Q	Total revenue	Average revenue
25	$23.5 \times 25 = \$587.5$	$587.5 \div 25 = \$23.5$
26	$23.5 \times 26 = \$611.0$	$611.0 \div 26 = \$23.5$

For the extra box, $\Delta Q = 1$:

$$\text{MR} = \frac{611.0 - 587.5}{1} = \$23.5.$$

Finding

For a price taker, $P = \text{AR} = \text{MR}$. Selling more does not lower the price of the boxes already sold.

Marginal cost and the change in profit

Definition: Marginal cost
The extra cost per unit of output: $\text{MC} = \Delta\text{TC}/\Delta Q$.

For a one-box increase, $\Delta Q = 1$:

$$\Delta\pi = \Delta\text{TR} - \Delta\text{TC} = \text{MR} - \text{MC}.$$

- If $\text{MR} > \text{MC}$, the extra box raises profit. Make it.
- If $\text{MR} < \text{MC}$, the extra box lowers profit. Do not make it.
- If $\text{MR} = \text{MC}$, the extra box leaves profit unchanged.

Best output at a price of \$23.5

Q	TC
24	\$350.1
25	\$372.5
26	\$397.1

From the printer's TC formula.
For 25 boxes:
$312.5 - 250.0 + 150.0 + 160.0$
$= \$372.5$.

MR $= P = \$23.5$. Each move has $\Delta Q = 1$.

- 25th box: MC $= (372.5 - 350.1)/1 = \$22.4$. Profit rises by $23.5 - 22.4 = \$1.1$.
- 26th box: MC $= (397.1 - 372.5)/1 = \$24.6$. Profit falls by $24.6 - 23.5 = \$1.1$.
- Best output: 25 boxes.

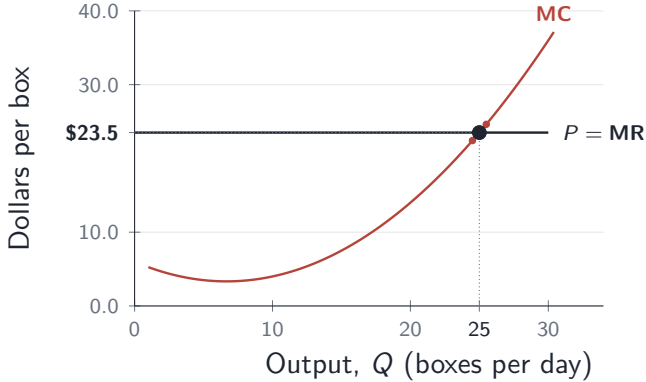
The output rule on a smooth curve

Smooth MC: $\text{MC} = 0.06Q^2 - 0.8Q + 6$.

The table chooses the last worthwhile whole box.

On a smooth curve, choose MR = MC where MC is rising.

At 25 boxes:
$0.06(25)^2 - 0.8(25) + 6$
$= 37.5 - 20.0 + 6.0 = \$23.5$.



Small dots show one-box MC at the midpoint of each move. The smooth curve closely follows them.

Fixed cost and the best output

Suppose unavoidable fixed cost rises by an amount F, at the same market price. For a one-box increase, $\Delta Q = 1$.

- Every TC rises by the same amount: new $\mathrm{TC}(Q) =$ old $\mathrm{TC}(Q) + F$.
- The extra fixed cost cancels when comparing two outputs:

$$\text{new MC} = \frac{[\mathrm{TC}(Q+1) + F] - [\mathrm{TC}(Q) + F]}{1} = \text{old MC}.$$

- Every MC is unchanged. MR is unchanged because the price is unchanged.

Finding

The profit-maximizing output is unchanged. Profit falls by F at every output, including zero. The long-run decision to stay can change.

Your turn: output at a price of $7.5

The same printer takes a market price of $7.5 per box.

1. What is MR for one more box?
2. What is MC for the 15th box? For the 16th?
3. Which output gives the highest profit in the table?

Each move is one box: $\Delta Q = 1$.

Q	TC
14	$220.5
15	$227.5
16	$235.5

Solution: output at a price of \$7.5

1. MR $= P =$ \$7.5.
2. 15th box: MC $= (227.5 - 220.5)/1 =$ \$7.0. Profit rises by $7.5 - 7.0 =$ \$0.5.
3. 16th box: MC $= (235.5 - 227.5)/1 =$ \$8.0. Profit falls by $8.0 - 7.5 =$ \$0.5.
4. Choose 15 boxes. Making fewer or more reduces profit.

Finding

Finding the best output does not establish whether the printer makes a profit. That requires total cost.

Part 3

Profit and Loss on the Graph

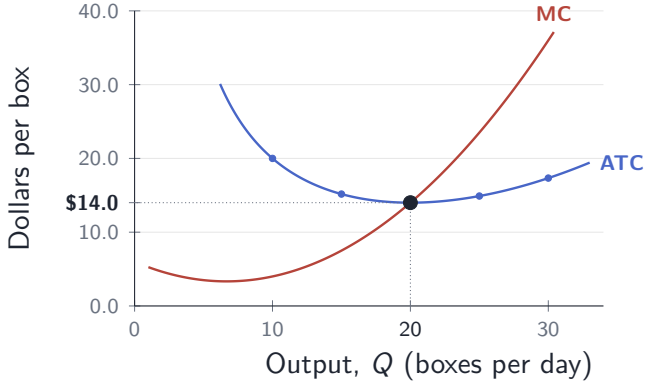
The printer's cost curves

ATC $=$ TC $\div$ Q. Smooth MC: $0.06Q^2 - 0.8Q + 6$.

Q	TC	ATC
10	\$200.0	\$20.0
15	\$227.5	\$15.2
20	\$280.0	\$14.0
25	\$372.5	\$14.9
30	\$520.0	\$17.3

At 20 boxes:
ATC $=$ 280.0 $\div$ 20
$=$ \$14.0.

MC crosses ATC at its lowest point.



Dots show the table values. Dollar amounts are rounded to one decimal place.

Profit per box and total profit

Definition: Profit from average total cost

For positive output,

$$\pi = \text{TR} - \text{TC} = (P \times Q) - (\text{ATC} \times Q) = (P - \text{ATC}) \times Q.$$

Profit per box times the number of boxes.

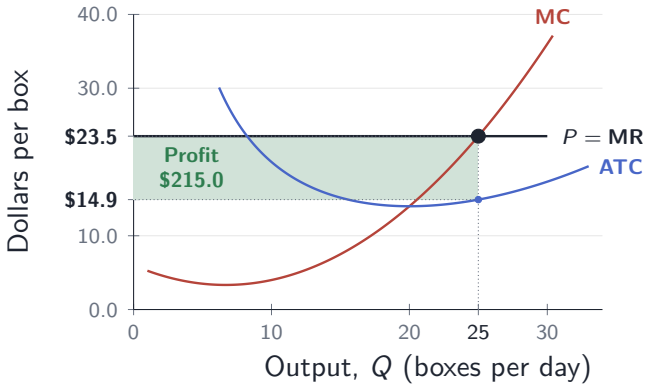
- Find output using MR and MC. Read ATC at that same output.
- $P > \text{ATC}$: profit. $P < \text{ATC}$: loss. $P = \text{ATC}$: zero profit.
- A rectangle of height P and width Q is TR.
- A rectangle of height ATC and width Q is TC. Their difference is profit.

Profit at a price of \$23.5

$P = \text{MR}$ meets rising MC at 25 boxes.

$\text{ATC} = 372.5 \div 25$
$= \$14.9.$

Height: $23.5 - 14.9 = \$8.6.$
Width: 25 boxes.
Profit: $8.6 \times 25 = \$215.0.$



Finding

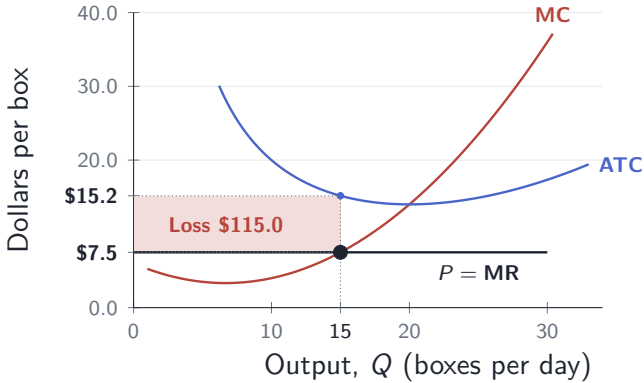
Check: $\text{TR} - \text{TC} = 23.5 \times 25 - 372.5 = 587.5 - 372.5 = \$215.0.$

Loss at a price of \$7.5

Best output: 15 boxes.

ATC $= 227.5 \div 15$
$\approx$ \$15.2, above $P =$ \$7.5.

Loss = height $\times$ width:
$[(227.5 \div 15) - 7.5] \times 15$
$= 227.5 - 112.5 =$ \$115.0.



Finding

Use unrounded ATC in the area calculation. MR = MC finds the best output, which can still make a loss.

Your turn: profit at a price of $14.0

The market price is $14.0 per box. Each move has $\Delta Q = 1$.

1. What is MC for the 20th box? For the 21st?
2. What is the best output?
3. What is ATC at that output?
4. What is economic profit?

Q	TC
19	$266.8
20	$280.0
21	$294.8

Solution: profit at a price of \$14.0

1. 20th box: $(280.0 - 266.8)/1 = \$13.2$. 21st box: $(294.8 - 280.0)/1 = \$14.8$.
2. Choose 20 boxes. The 20th adds to profit. The 21st reduces it.
3. ATC $= 280.0 \div 20 = \$14.0$.
4. $\pi = (14.0 - 14.0) \times 20 = \$0.0$. Revenue equals cost: $14.0 \times 20 = \$280.0 = \text{TC}$.

Part 4

Shutdown in the Short Run, Exit in the Long Run

Shutdown, exit and sunk cost

Definition: Shutdown

A short-run decision to produce nothing for now. The firm still pays its unavoidable fixed costs.

Definition: Exit

A long-run decision to leave the market. The firm avoids the future costs of staying, including lease payments once the commitment ends.

Definition: Sunk cost

A cost already incurred or irrevocably committed that cannot be recovered. It does not affect what the firm should do next.

The printer: produce or shut down?

At a price of \$7.5, the best positive output is 15 boxes.

- Revenue: $7.5 \times 15 = \$112.5$. Variable cost: $227.5 - 160.0 = \$67.5$.
- Produce: $\pi = 112.5 - 67.5 - 160.0 = -\$115.0$.
- Shut down: $\pi = 0.0 - 0.0 - 160.0 = -\$160.0$.
- Producing covers VC and puts $112.5 - 67.5 = \$45.0$ toward fixed cost.

Finding

Keep producing. The loss is smaller by $160.0 - 115.0 = \$45.0$.

The shutdown rule

Compare the best positive output with producing nothing.

- Producing gives TR $-$ VC $-$ FC. Shutting down gives $-$FC.
- Producing adds TR $-$ VC. Fixed cost cancels.
- For $Q > 0$, divide TR and VC by output: TR $\div Q = P$ and VC $\div Q =$ AVC.

Finding

Shut down if TR $<$ VC, or $P <$ AVC.
Continue if TR $>$ VC. If TR $=$ VC, producing and shutting give the same profit.

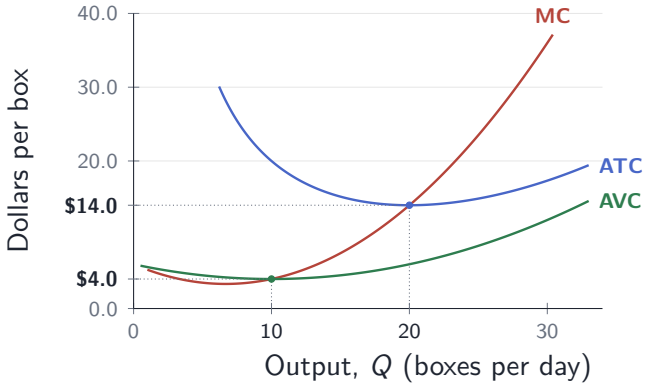
Shutdown and break-even on the graph

$\text{AVC} = (\text{TC} - 160) \div Q = 4 + (Q - 10)^2/50$; $\text{ATC} = \text{AVC} + 160/Q$.

Shutdown price:
minimum AVC at 10 boxes:
$40.0 \div 10 = \$4.0$.

Break-even price:
minimum ATC at 20 boxes:
$280.0 \div 20 = \$14.0$.

MC crosses each average-cost curve at its minimum.



Three price regions

Finding

- $P < \$4.0$: shut down now. Price cannot cover AVC at any output.
- $\$4.0 < P < \$14.0$: produce at a loss in the short run. Exit in the long run if these conditions persist.
- $P > \$14.0$: produce and make an economic profit.
- At \$4.0, producing and shutting give the same result. At \$14.0, economic profit is zero.

In the long run, compare revenue with all avoidable economic costs. Exit if TR $<$ TC, or $P <$ ATC at the best output.

An unavoidable lease

- The printer's lease must be paid until it expires, even if production stops.
- Revenue covers paper, ink and other avoidable operating costs, but not total cost.
- Continue producing while the lease is unavoidable. Production reduces the loss.
- When the lease expires, renewal becomes a cost the printer can avoid.
- If revenue still cannot cover all costs, do not renew. Exit the market.

Your turn: the shutdown price

The printer's minimum AVC is \$4.0 at 10 boxes. At that output, VC is \$40.0 and fixed cost is \$160.0.

1. At a price of \$4.0, what is the profit from making 10 boxes?
2. What is the profit from shutting down?
3. Which choice is better at \$4.0? What if the price falls below \$4.0?

Solution: the shutdown price

1. Produce 10 boxes: $4.0 \times 10 - 40.0 - 160.0 = -\$160.0$.
2. Shut down: $0.0 - 0.0 - 160.0 = -\$160.0$.
3. At \$4.0 the choices give the same profit. Below \$4.0, shut down: no output covers variable cost.

Finding

The shutdown price is minimum AVC. It is not the price needed to cover total cost.

Part 5

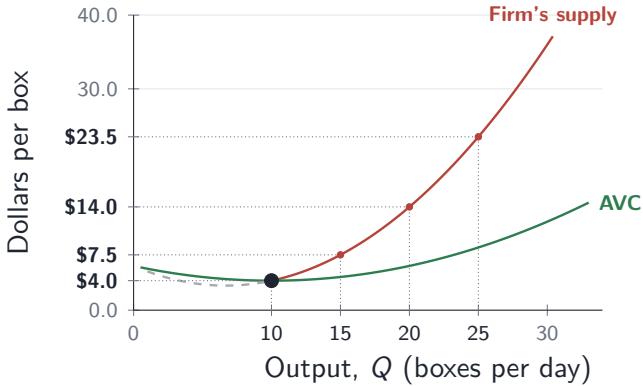
The Short-Run Supply Curve

The printer's supply schedule and curve

Price	Boxes
Below \$4.0	0
\$7.5	15
\$14.0	20
\$23.5	25

The rising MC curve gives output at each price, provided producing covers variable cost.

Below \$4.0, supply is zero.



The firm's short-run supply curve

Finding

The firm's positive-output short-run supply curve is the rising part of MC at and above minimum AVC. Below minimum AVC, the firm supplies zero.

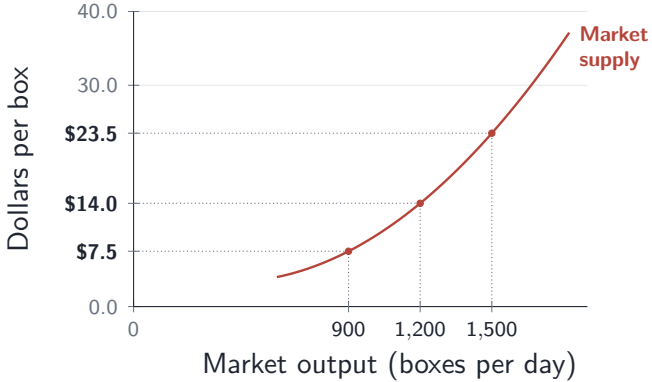
- Why MC: the firm chooses output where $P = \text{MR} = \text{MC}$.
- Why the AVC threshold: below it, revenue cannot cover variable cost.
- At minimum AVC, either zero output or the shutdown-point output is optimal.
- ATC measures profit. It does not set the short-run supply threshold.

From one firm to market supply

60 identical printers.

Price	Market output
\$7.5	$60 \times 15 = 900$
\$14.0	$60 \times 20 = 1{,}200$
\$23.5	$60 \times 25 = 1{,}500$

The number of firms is fixed in the short run. A higher price raises each firm's output.



Definition: Market supply

The horizontal sum of the firms' supply curves. At each price, add the quantities supplied by all firms.

Part 6

Entry and Exit

Entry and exit change market supply

Assume identical firms, free entry and exit, and unchanged input prices.

Positive economic profit

- New firms enter.
- Market supply shifts right.
- The price falls and profit shrinks.

Persistent economic losses

- Some firms exit.
- Market supply shifts left.
- The price rises and losses shrink.

Finding

Entry and exit stop at zero economic profit: $P =$ minimum ATC. For the printer, $(14.0 - 14.0) \times 20 = \$0.0$.

Zero economic profit

Definition: Zero economic profit

Total revenue covers all economic costs, including the opportunity costs of the owner's time, capital and other resources.

- The salary given up to run the printing business is part of economic cost.
- Accounting profit can be positive when economic profit is zero.
- The owner earns the normal return available elsewhere. There is no extra economic profit to attract entry.

Part 7

Firm and Industry Equilibrium, and Changes to Equilibrium

Short-run and long-run equilibrium

Definition: Short-run equilibrium

Market quantity demanded equals quantity supplied. Each firm maximizes profit at the market price. Firms can make profits or losses.

Definition: Long-run equilibrium

With identical firms, free entry and exit, and unchanged input prices: firms earn zero economic profit and produce at efficient scale.

Definition: Efficient scale

The output at which average total cost is lowest.

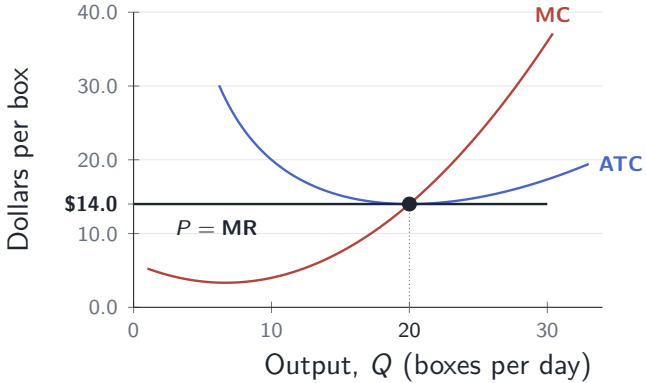
The firm in long-run equilibrium

$P = \text{MR}$: price taking.

$\text{MR} = \text{MC}$: the output rule.

$P = \text{ATC}$: zero economic profit.

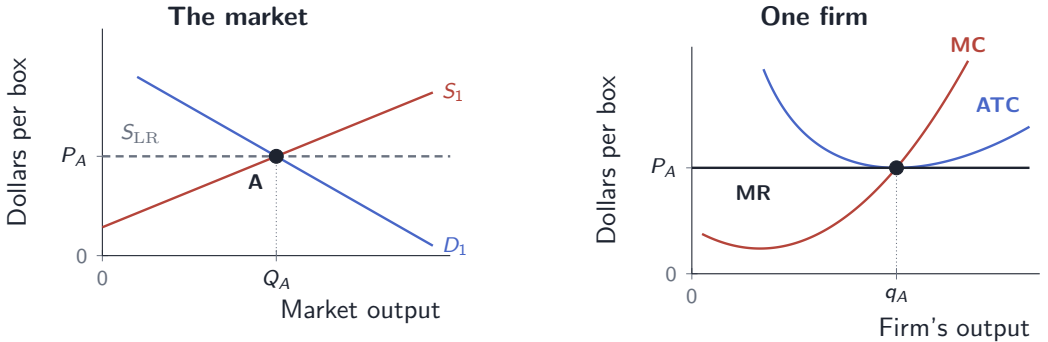
MC meets ATC at its minimum: the efficient scale.



Finding

$P = \text{MR} = \text{MC} = \text{minimum ATC} = \$14.0$.
At 20 boxes, $\text{TR} = 14.0 \times 20 = \$280.0 = \text{TC}$.

Market and firm at A

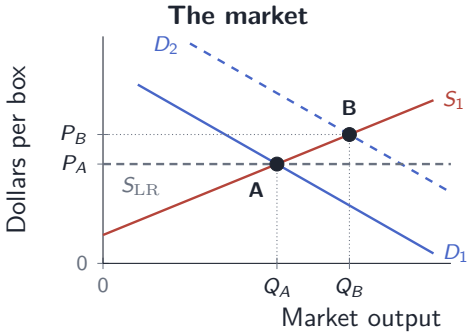


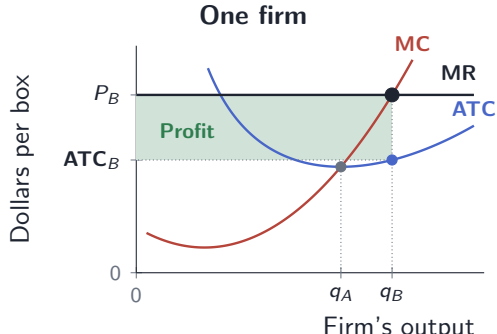
Finding

At A, price equals minimum ATC. Each firm produces at efficient scale and earns zero economic profit. S_{LR} is long-run market supply.

These are schematic panels: compare the direction of changes, not their size.

Market and firm at B

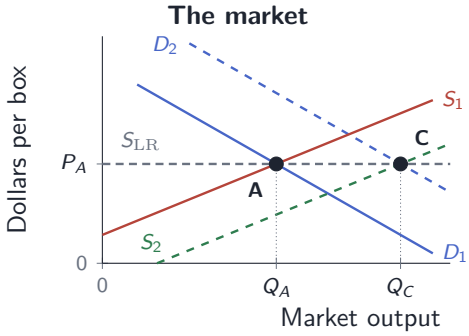


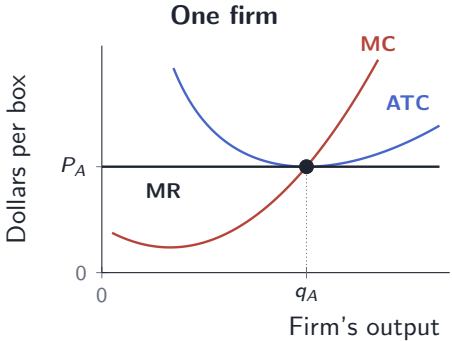


Finding

Demand rises. With the number of firms fixed, price and each firm's output rise.
Profit rectangle: height $P_B - \text{ATC}_B$; width q_B; area $(P_B - \text{ATC}_B) \times q_B$.

Market and firm at C





Finding

Profit attracts entry: supply shifts from S_1 to S_2. Price returns to P_A.
Each firm returns to q_A and zero economic profit. Market output rises from Q_A to Q_C because there are more firms.

A fall in market demand

- In the short run, demand shifts left and the market price falls.
- Each operating firm reduces output along its MC curve.
- All, some, or no firms shut down: this depends on whether the new price falls below their minimum AVC.
- If losses persist, some firms exit. Market supply shifts left and price rises.

Finding

With identical firms and unchanged input prices, exit returns price to minimum ATC. There are fewer firms, each at its original efficient scale.

When long-run supply slopes up

- A scarce input: industry expansion bids up its price. Firms' costs rise. More output requires a higher market price.
- Different costs: the lowest-cost firms enter first. More output requires higher-cost firms, which enter only at a higher price.

Finding

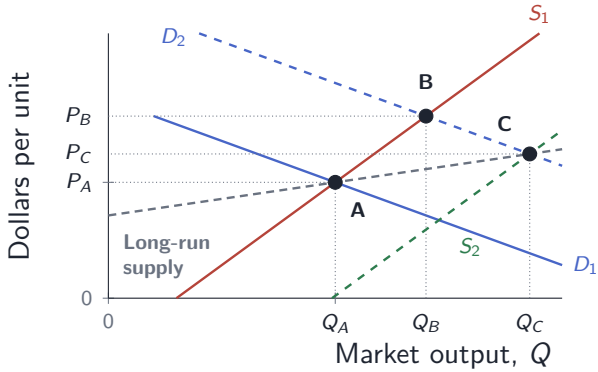
Free entry does not require horizontal long-run market supply. Long-run supply can slope upward.

An upward-sloping long-run supply curve

A: the initial long-run equilibrium.

B: demand rises. With the number of firms fixed, price rises along S_1.

C: entry shifts supply to S_2. Costs have risen, so price remains above P_A.



Finding

Entry lowers price from P_B to P_C. Higher costs keep P_C above P_A.

The marginal firm

Definition: Marginal firm

The firm that would exit in the long run if the price fell any lower. At the current price it earns zero economic profit.

- When costs differ, the market price equals the marginal firm's minimum ATC.
- Lower-cost firms can retain an economic profit if entrants cannot copy their cost advantage.
- Entry stops when no remaining entrant can earn a positive economic profit.

Why competition is efficient

- The market price measures what the last unit is worth to buyers.
- Each firm chooses $P = \text{MC}$: the last unit's value equals its cost.
- Producing more would cost more than buyers are willing to pay. Producing less would give up worthwhile trades.

Finding

With no external costs or benefits, the competitive market produces the efficient quantity.

Review

Practice and Summary

Practice 1: the firm's demand curve

Which change moves a competitive printer's horizontal demand curve upward?

(A) The market price rises.

(B) The printer's fixed cost rises.

(C) The printer increases its own output.

(D) The printer's average total cost rises.

Practice 1: answer

Answer: (A) The market price rises.

Finding

The firm's demand curve is horizontal at the market price. A higher market price moves that line upward.

- A cost change affects profit or supply, not the price buyers will pay this firm.
- This firm's output is too small to move the market price.

Practice 2: a table of total costs

The price is \$14.0 per box. Using the table, what happens to daily profit if the printer increases output from 20 to 21 boxes?

(A) It rises by \$14.0.

(B) It falls by \$14.8.

(C) It falls by \$0.8.

(D) It is unchanged.

Q	TC
20	\$280.0
21	\$294.8

Practice 2: answer

Answer: (C) Profit falls by \$0.8.

1. One extra box: $\Delta Q = 1$ and $\text{MR} = \$14.0$.
2. $\text{MC} = (294.8 - 280.0)/1 = \$14.8$.
3. $\Delta\pi = \text{MR} - \text{MC} = 14.0 - 14.8 = -\$0.8$.

Finding

The extra cost exceeds the extra revenue. Total revenue rises, but profit falls.

Practice 3: a cost rectangle

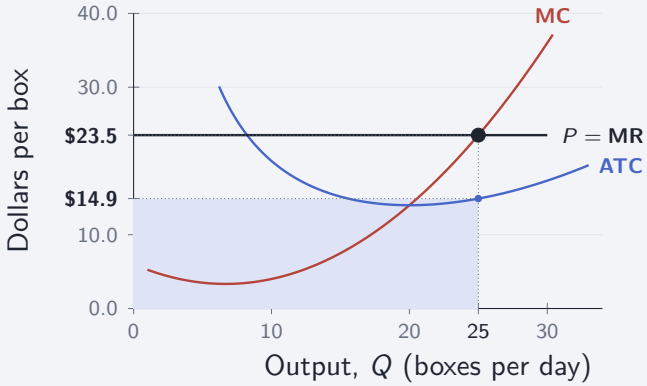
The printer makes 25 boxes at a price of \$23.5. What does the shaded rectangle measure?

(A) Total revenue.

(B) Total cost.

(C) Economic profit.

(D) Variable cost.



Practice 3: answer

Answer: (B) Total cost.

- Height $=$ ATC $=$ \$14.9. Width $=$ 25 boxes.
- Area $= 14.9 \times 25 =$ \$372.5 $=$ TC.
- Total revenue is the taller rectangle: $23.5 \times 25 =$ \$587.5.
- The difference is profit: $587.5 - 372.5 =$ \$215.0.

Practice 4: a loss-making printer

At a price of \$7.5, the printer makes 15 boxes and loses \$115.0 per day. Its fixed cost of \$160.0 is unavoidable today.

> **Finding**
> Shutting down today would reduce its loss.

True or false?

Practice 4: answer

Answer: False.

- Producing loses \$115.0. Shutting down loses \$160.0.
- Producing reduces the loss by $160.0 - 115.0 = \$45.0$.
- Revenue covers variable cost and contributes to fixed cost.

Finding

A loss alone does not imply shutdown. Compare the loss from producing with the loss from producing nothing.

Practice 5: a lasting increase in demand

A competitive market starts in long-run equilibrium. Firms have identical costs, entry is free, and input prices do not change as the industry grows.

> **Finding**
> After a lasting increase in demand, long-run market output is higher even though each firm returns to its original output.

True or false?

Practice 5: answer

Answer: True.

1. Higher demand initially raises price, each firm's output and profit.
2. Profit attracts entry. More firms shift market supply to the right.
3. Price returns to minimum ATC. Each firm returns to efficient scale.
4. There are more firms, so market output is higher.

Which rule answers which question

Question	Rule
What price does one firm face?	The market price: $P = \text{AR} = \text{MR}$.
How much should it produce?	$\text{MR} = \text{MC}$ on rising MC, if it operates.
Does it make a profit?	$\pi = (P - \text{ATC}) \times Q$.
Should it shut down now?	Shut if $P < \text{AVC}$; threshold: minimum AVC.
Should it stay in the long run?	Exit if revenue cannot cover all avoidable costs.
What changes market supply?	Entry shifts it right; exit shifts it left.
Where does entry and exit stop?	$P =$ minimum ATC with identical firms and unchanged input prices.